

Segmentation of 3D Gaussians using Masked Gradients

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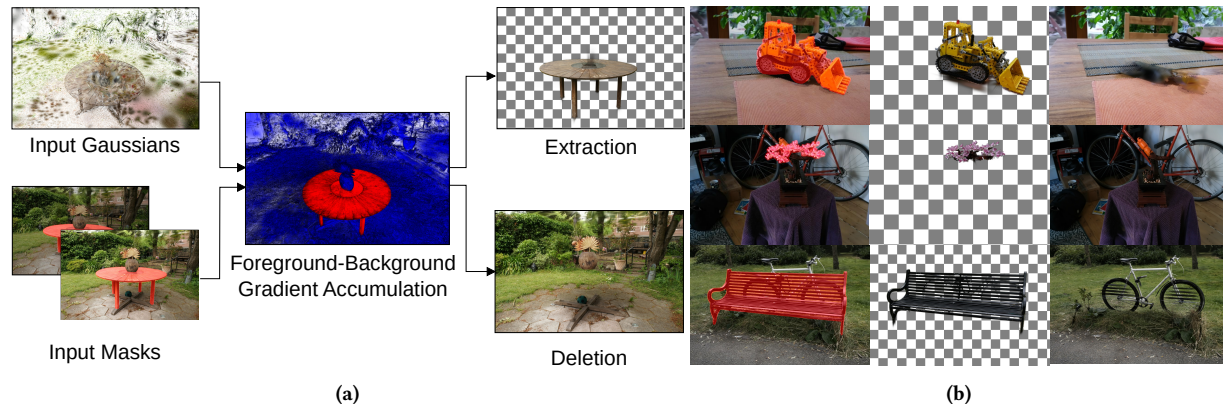


Figure 1: (a) Pipeline: The process begins with a Gaussian splatting model and 2D segmentation masks as input. Gradients are computed for pixels with respect to the color of each Gaussian, and these gradients are accumulated. A threshold is then applied to generate a 3D mask that isolates the relevant Gaussians corresponding to the segmented regions. **(b) Example 3D Segmentations:** From left to right—input mask, extraction, and deletion results. The last row illustrates a limitation of the method, which requires highly accurate 2D masks for optimal performance. Zooming in is recommended for better clarity.

Abstract

We present a simple yet powerful method for segmenting 3D Gaussian splatting models using 2D segmentation masks from a limited number of viewpoints. Our approach leverages inference-time gradient backpropagation, making it both easy to implement and fast enough for interactive segmentation. The underlying formulation ensures high accuracy, making this method a highly effective tool for 3D segmentation and related downstream tasks such as AR, VR, 3DGS editing, and digital twins.

Keywords

3D Gaussian Splatting, 3D Segmentation

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1 Introduction

3D Gaussian splatting (3DGS) is a popular technique for rendering novel viewpoints using 3D Gaussian distributions as rendering primitives[Kerbl et al. 2023]. It is particularly known for its speed and flexibility in editing. Beyond color information, 3DGS can be extended to learn 3D feature fields, enabling applications in 2D and 3D segmentation and localization tasks[Qin et al. 2023; Zhou et al. 2024].

However, while feature fields excel in localization and 2D segmentation [Qin et al. 2023], they often struggle with clean 3D segmentation, leading to artefacts like floaters. Additionally, training these models is slow and computationally expensive.

Our approach addresses these challenges by using 2D segmentation masks generated by foundation models and applying inference-time gradient backpropagation. Inspired by 2D techniques for visualization, such as DeepDream[dee [n. d.]] and Grad-CAM[Selvaraju et al. 2019], our method identifies Gaussians responsible for specific 2D masks, enabling more accurate and efficient 3D segmentation.

2 Method

3D Gaussian Splatting is a rendering technique that uses Gaussian distributions as primitives[Kerbl et al. 2023]. Each Gaussian is characterized by 3D distribution parameters (mean location and covariance) and has associated anisotropic color and opacity. The Gaussians are sorted by their perpendicular distance from the camera, and the 3D distribution is projected onto a 2D plane using the Jacobian of the projection transformation[Zwicker et al. 2001]. The span of each Gaussian on the 2D plane represents the region it

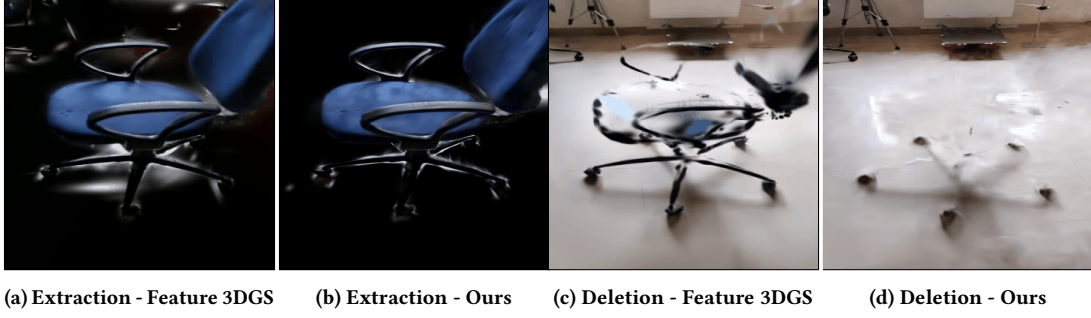


Figure 2: Comparison of our approach with feature 3DGS. Our approach is cleaner and more accurate.

influences, with opacity decreasing exponentially from its centre. After the 3D-to-2D transformation, alpha compositing combines the contributions from all Gaussians to render the final image.

Consider the color C of a pixel at (x, y) in a 3DGS rendering,

$$C = \sum_{n \leq N} c_n \alpha_n \prod_{m < n} (1 - \alpha_m) \quad (1)$$

$$= \sum_{n \leq N} c_n \alpha_n T_n \quad (2)$$

Where N is the total number of Gaussians, each indexed by its sorted position, c_n is the color associated with the n th Gaussian, α_n is the opacity of the n th Gaussian at (x, y) adjusted with exponential falloff, and $T_n = \prod_{m < n} (1 - \alpha_m)$ is the transmittance of n th Gaussian at (x, y) .

Taking the derivative with respect to color of k th Gaussian c_k ,

$$\frac{\partial C}{\partial c_k} = \alpha_k T_k \quad (3)$$

This value is zero only if either the transmittance or opacity is zero, indicating that the Gaussian does not contribute to the final color. A non-zero gradient means the Gaussian influences the pixel color.

Our approach identifies the gradients of Gaussians that influence the 2D segmentation regions. We maintain a buffer corresponding to the number of Gaussians, accumulating gradients for each mask: adding gradients for the foreground and subtracting them for the background (inverted mask). Finally, 3D Gaussians with positive gradients in the buffer are segmented out as a pre-pass for downstream tasks.

3 Experiment and Results

We compared object extraction and deletion using our method against Feature 3DGS - a feature field based 3D segmentation method (figure 2). Feature 3DGS utilized text embeddings from prompts to find Gaussians with similar features. For gradient-based segmentation, we selected specific viewpoints and used YOLO-world[Cheng et al. 2024] to identify the initial bounding box, followed by SAM 2[Ravi et al. 2024] to estimate and track masks.

Our results show that our method outperforms feature field-based segmentation in accuracy. While feature field-based segmentation is faster during inference, it is slow and computationally expensive to train. In contrast, our approach works even with vanilla 3DGS, resulting in faster training. Despite the need to select multiple frames during inference, our method remains sufficiently fast

for interactive applications - 27 seconds for segmenting one scene (from Mip-NeRF 360) including mask generation and 3.5 seconds excluding mask generation.

4 Conclusion

We presented a method that uses inference-time backpropagation to segment 3D Gaussian splatting models. While it relies on the accuracy of 2D masks—generated interactively or automatically—experiments show that it achieves higher accuracy than feature field-based 3D segmentation.

The method requires multiple forward and backward passes during inference, which slows down inference but eliminates the need for feature field training, resulting in faster setup and lower memory usage. Its strong mathematical basis ensures clean segmentation with sufficient viewpoints and accurate masks.

Acknowledgments

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